

HANDS-ON ACTIVITY FOCUS GUIDE

The Human Minus Sign

Turn raw joystick Y values into motor-power requests

20 MIN ACTIVITY	8 STUDENT ROLES	MATH + MOTION FORMAT	1 SAFETY LEAD
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ACTIVITY OUTCOME Students read assignment statements from right to left, negate FTC Y-axis values, and connect the stored result to a predicted tank-drive command.

Materials

- Role cards: RAW VALUE, MINUS SIGN, POWER VARIABLE, and MOTION PREDICTOR.
- Number cards: -1.0, -0.7, -0.2, 0.0, +0.35, +0.8, and +1.0.
- Printed joystick-conversion sheet from this guide.
- BasicTwoMotorTeleOp.java displayed at the leftPower/rightPower assignments.
- Optional secured robot with power telemetry visible.

Before students enter

- State the lesson convention: positive stored power is intended as robot-forward after direction configuration.
- Keep raw gamepad values separate from stored power values by using different colored cards.
- Prepare both single-value and left/right tank-drive examples.
- If using the robot, secure it and keep it disabled until the final extension.

SAFETY BOUNDARY Students never cross into the robot test area while the OpMode is enabled. STOP and visually confirm DISABLED before anyone approaches.

Student setup and roles

Assign one primary role to each student. Rotate roles on a second trial if time permits. The safety lead may stop the activity at any time.

STUDENT	ROLE	RESPONSIBILITY
1	Left raw value	Displays gamepad1.left_stick_y.
2	Right raw value	Displays gamepad1.right_stick_y.
3	Left minus sign	Changes the sign of the left raw value.
4	Right minus sign	Changes the sign of the right raw value.
5	Left power	Holds the result stored in leftPower.
6	Right power	Holds the result stored in rightPower.
7	Motion predictor	Predicts forward, backward, stop, curve, or spin.
8	Recorder	Writes raw values, stored values, and predictions.

Ground rules

- Predict before testing; record evidence before proposing a correction.
- Only the Driver Station operator touches INIT, PLAY, or STOP.
- Any student may call STOP for unexpected motion, instability, heat, noise, or an unclear boundary.
- Change only one variable between tests so the evidence remains interpretable.

POWERING ON gate

SAY AND DO ALL SIX STEPS 1. Secure the robot. 2. Confirm DISABLED. 3. Look around. 4. Make sure everyone is clear. 5. Call POWERING ON! and wait. 6. Visually confirm clear, then enable/PLAY.

After every test: press STOP, visually confirm DISABLED, and only then permit anyone to approach the robot.

20-minute teacher walkthrough

1. Read assignment right to left | 4 minutes

TEACHER	<i>"Display double leftPower = -gamepad1.left_stick_y;. Start at the right side and narrate the order of work."</i>
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STUDENT	<i>"Students identify the raw input, apply unary minus, and place the result into leftPower. They identify double as a decimal-number type."</i>
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ASK	<i>"Which side is evaluated first?"</i>
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LISTEN FOR The expression on the right; its result is assigned to the variable on the left.

TEACHING NOTE The equals sign means assignment here, not a question asking whether two things are equal.

2. Become the sign changer | 5 minutes

TEACHER	<i>"Hand RAW VALUE -0.7 to a student. Ask MINUS SIGN to transform it without changing its magnitude."</i>
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STUDENT	<i>"RAW VALUE shows -0.7; MINUS SIGN announces negative of negative 0.7; POWER VARIABLE displays +0.7."</i>
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ASK	<i>"What happens to 0.0?"</i>
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LISTEN FOR Its negation is still 0.0; zero has no forward/backward direction.

TEACHING NOTE Say negate or change the sign, not make it negative. A positive raw value becomes negative.

3. Process a tank-drive pair | 6 minutes

TEACHER	<i>"Give the left and right raw students different cards. Both paths operate at the same time conceptually, but calculate each path clearly."</i>
STUDENT	<i>"Each minus-sign student transforms a raw value; power students display results; the motion predictor compares left and right."</i>
ASK	<i>"If stored powers are +0.7 and +0.3, what is predicted?"</i>
LISTEN FOR Both sides request forward, but the left side is stronger, so the robot should curve toward the slower right side.	
TEACHING NOTE Physical direction still depends on correct motor mapping and direction configuration.	

4. Separate zero power from STOP | 2 minutes

TEACHER	<i>"Use raw 0.0 on both sticks. Ask whether the OpMode ends."</i>
STUDENT	<i>"Both power students show 0.0. The predictor says motors are commanded to stop, but the loop remains active."</i>
ASK	<i>"What action actually ends TeleOp?"</i>
LISTEN FOR The driver presses STOP, making the active condition false.	

5. Optional telemetry check | 3 minutes

TEACHER	<i>"Secure the robot and complete the POWERING ON sequence. Move one stick gently and compare raw direction, stored sign, and telemetry."</i>
STUDENT	<i>"Students state the expected sign before reading the telemetry, then STOP and confirm DISABLED."</i>
ASK	<i>"Why may the released stick show a tiny nonzero value?"</i>
LISTEN FOR Real joysticks can drift near center; a later program can apply a deadband.	
TEACHING NOTE Do not introduce deadband code until students can trace the original expression.	

Joystick sign-conversion practice

Student/team: _____ Date: _____ Robot: _____

RAW Y VALUE	APPLY -Y	STORED POWER	PREDICTED REQUEST
-1.0	$-(-1.0)$		
-0.7	$-(-0.7)$		
0.0	$-(0.0)$		
+0.35	$-(+0.35)$		
+1.0	$-(+1.0)$		
Student value: ____			
STUDENT EVIDENCE RULE Write what was commanded and what was observed. Do not write a correction until STOP has been pressed and DISABLED has been confirmed.			

Safety sign-off

Before live testing, the safety lead confirms: robot secure ☐ DISABLED ☐ boundary clear ☐ POWERING ON called and wait completed ☐

After testing: STOP pressed ☐ DISABLED visually confirmed ☐ Safety lead initials: _____

COMPLETION CRITERIA The student uses lesson vocabulary, predicts before testing, records evidence, explains one result from the code or math, and follows STOP/DISABLED rules before approaching the robot.

Debrief

ASK	LISTEN FOR
Why is Y negated?	FTC gamepad Y is normally negative when pushed forward; negation creates a convenient positive-forward command.
What does double communicate?	The variable stores a floating-point/decimal numeric value.
Are leftPower and rightPower permanent fields here?	No. They are local variables created during each loop iteration.
Does +0.5 telemetry guarantee physical forward motion?	No. It is the command; physical direction also depends on configuration and mounting.

Troubleshooting without guessing

OBSERVATION	NEXT CHECK AFTER STOP/DISABLED
Student says minus always makes negative	Use +0.4 and -0.4 side by side; unary minus reverses either sign.
Student reads assignment left to right	Cover the left side, calculate the right expression, then reveal the destination.
Predicted curve is reversed	Identify which side is slower; a differential drive curves toward the slower side.
Telemetry is slightly nonzero at center	Recognize gamepad drift; note deadband as a later improvement.

Technical notes for the teacher

- FTC joystick axes typically report floating-point values from -1.0 to +1.0.
- Pushing a Y axis forward normally produces a negative raw value.
- Unary minus is applied before the result is assigned to the power variable.
- The power variables in this program are recreated on every loop iteration.
- Negating input does not replace correct motor direction configuration.

EXIT TICKET If right_stick_y is +0.35, what is stored in rightPower, and what motion does that side request under the lesson convention?